Efficiency and resource measurement

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One complete model comparison record.

Preserve raw timings, quality counts, resource scope and honest limits.

Keep one habit: Recall → W1–W3/W5 → independent comparison → project protocol → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Dataset shift in neural models. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Sequence representations. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- A neural model can be faster yet fail the task's accuracy requirement. **Training** cost and **prediction** cost are different: **training** updates parameters, while inference uses the fitted model.

  Compare quality and resources together under a declared test, rather than calling a model efficient from one small number.

- Use these invented classroom timings for models A and B. Each timed pass **process**es the same 100 pre**process**ed examples sequentially, one at a time, on the same device and precision.

  Start before the first **prediction** and stop after the final **prediction** completes. Loading and pre**process**ing are excluded for both. One untimed warm-up pass precedes three timed passes; the warm-up work must still be reported.

- The validation cases are the same for both models and are scored once each; repeating timing passes does not create 300 independent quality examples. No actual benchmark was run to produce this table.

Dataset shift in neural models

> **My remembered idea, example and correction**

Transfer learning

My remembered idea, example and correction

Sequence representations

My remembered idea, example and correction

Lesson reference for workbook practice

Measurement	Model A	Model B
Pass 1 seconds	2.0	1.2
Pass 2 seconds	2.2	1.1
Pass 3 seconds	1.8	1.3
Correct on 100 validation cases	92	90

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main invented timings on the same 100 examples: A [2.0, 2.2, 1.8] seconds, 92 correct; B [1.2, 1.1, 1.3], 90 correct. Same device, precision and sequential batch-one boundary; loading/preprocessing excluded. One untimed warm-up then three timed passes. Medians 2.0 and 1.2 seconds; amortised times 20 and 12 ms; median-pass throughput 50 and about 83.33 examples/second.

  Time reduction 40%; throughput increase about 66.67%. Required at least 91 correct and at most 25 ms: only A qualifies. Alternative requirement at least 90 correct set in advance: both qualify and B wins the lower-time rule. Three timing passes execute 300 predictions plus 100 warm-up predictions per model, but quality still uses 100 distinct cases.

  Dense 4→3→2: weights 12+6 = 18, biases 3+2 = 5, total 23, raw payload 92 bytes at four bytes each. This is not file size, peak memory, energy or carbon. Separate picture uses means over 40 cases: P .32/.40/.48 seconds, 38 correct; Q .20/.24/.28, 36 correct. Mean times .40/.24, amortised 10/6 ms, accuracy 95%/90%. At 93% and 12 ms only P qualifies; at 90% both qualify.

  No actual benchmark, latency distribution, energy or future accuracy is supplied.

- For A and B, calculate median pass time, amortised milliseconds per example and median-pass **throughput**. State the measured boundary and one **latency** claim these numbers cannot support.

My reasoning and evidence

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main invented timings on the same 100 examples: A [2.0, 2.2, 1.8] seconds, 92 correct; B [1.2, 1.1, 1.3], 90 correct. Same device, precision and sequential batch-one boundary; loading/preprocessing excluded. One untimed warm-up then three timed passes. Medians 2.0 and 1.2 seconds; amortised times 20 and 12 ms; median-pass throughput 50 and about 83.33 examples/second.

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  No actual benchmark, latency distribution, energy or future accuracy is supplied.

- With at least 91 correct out of 100 and at most 25 ms amortised time required, which model qualifies? Calculate B's time reduction relative to A and explain why the percentage **throughput** increase differs.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main invented timings on the same 100 examples: A [2.0, 2.2, 1.8] seconds, 92 correct; B [1.2, 1.1, 1.3], 90 correct. Same device, precision and sequential batch-one boundary; loading/preprocessing excluded. One untimed warm-up then three timed passes. Medians 2.0 and 1.2 seconds; amortised times 20 and 12 ms; median-pass throughput 50 and about 83.33 examples/second.

  Time reduction 40%; throughput increase about 66.67%. Required at least 91 correct and at most 25 ms: only A qualifies. Alternative requirement at least 90 correct set in advance: both qualify and B wins the lower-time rule. Three timing passes execute 300 predictions plus 100 warm-up predictions per model, but quality still uses 100 distinct cases.

  Dense 4→3→2: weights 12+6 = 18, biases 3+2 = 5, total 23, raw payload 92 bytes at four bytes each. This is not file size, peak memory, energy or carbon. Separate picture uses means over 40 cases: P .32/.40/.48 seconds, 38 correct; Q .20/.24/.28, 36 correct. Mean times .40/.24, amortised 10/6 ms, accuracy 95%/90%. At 93% and 12 ms only P qualifies; at 90% both qualify.

  No actual benchmark, latency distribution, energy or future accuracy is supplied.

- Count weights, biases and total parameters in the 4 to 3 to 2 network. Calculate raw parameter bytes at four bytes each. Explain why this cannot be reported as peak **training** memory or an energy measurement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main invented timings on the same 100 examples: A [2.0, 2.2, 1.8] seconds, 92 correct; B [1.2, 1.1, 1.3], 90 correct. Same device, precision and sequential batch-one boundary; loading/preprocessing excluded. One untimed warm-up then three timed passes. Medians 2.0 and 1.2 seconds; amortised times 20 and 12 ms; median-pass throughput 50 and about 83.33 examples/second.

  Time reduction 40%; throughput increase about 66.67%. Required at least 91 correct and at most 25 ms: only A qualifies. Alternative requirement at least 90 correct set in advance: both qualify and B wins the lower-time rule. Three timing passes execute 300 predictions plus 100 warm-up predictions per model, but quality still uses 100 distinct cases.

  Dense 4→3→2: weights 12+6 = 18, biases 3+2 = 5, total 23, raw payload 92 bytes at four bytes each. This is not file size, peak memory, energy or carbon. Separate picture uses means over 40 cases: P .32/.40/.48 seconds, 38 correct; Q .20/.24/.28, 36 correct. Mean times .40/.24, amortised 10/6 ms, accuracy 95%/90%. At 93% and 12 ms only P qualifies; at 90% both qualify.

  No actual benchmark, latency distribution, energy or future accuracy is supplied.

- A runs on one device with batch size 1; B runs on another with batch size 32. Only B's fastest trial and the winning **training** run are reported. Explain how to compare and report the results fairly.

> **My reasoning and evidence**
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MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- For 50 shared validation cases, P has times 0.6, 0.5, 0.4 seconds and 47 correct; Q has 0.3, 0.4, 0.5 seconds and 45 correct. Require at least 92 percent accuracy and at most 12 ms amortised time.

  Give median times, amortised times, **throughput**, accuracy and eligible models. A separate dense network has dimensions 3 to 2 to 1 with one bias per **output**: find its parameter count and raw bytes at four bytes each.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Evaluate a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled efficiency and resource measurement **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate **throughput** and amortised time per example from repeated timings.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Compare under requirements for Efficiency and resource measurement. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Efficiency and resource measurement. Proposed explanation: "Fewer parameters always means faster". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- For 50 shared validation cases, P has times 0.6, 0.5, 0.4 seconds and 47 correct; Q has 0.3, 0.4, 0.5 seconds and 45 correct. Require at least 92 percent accuracy and at most 12 ms amortised time.

  Give median times, amortised times, **throughput**, accuracy and eligible models. A separate dense network has dimensions 3 to 2 to 1 with one bias per **output**: find its parameter count and raw bytes at four bytes each.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

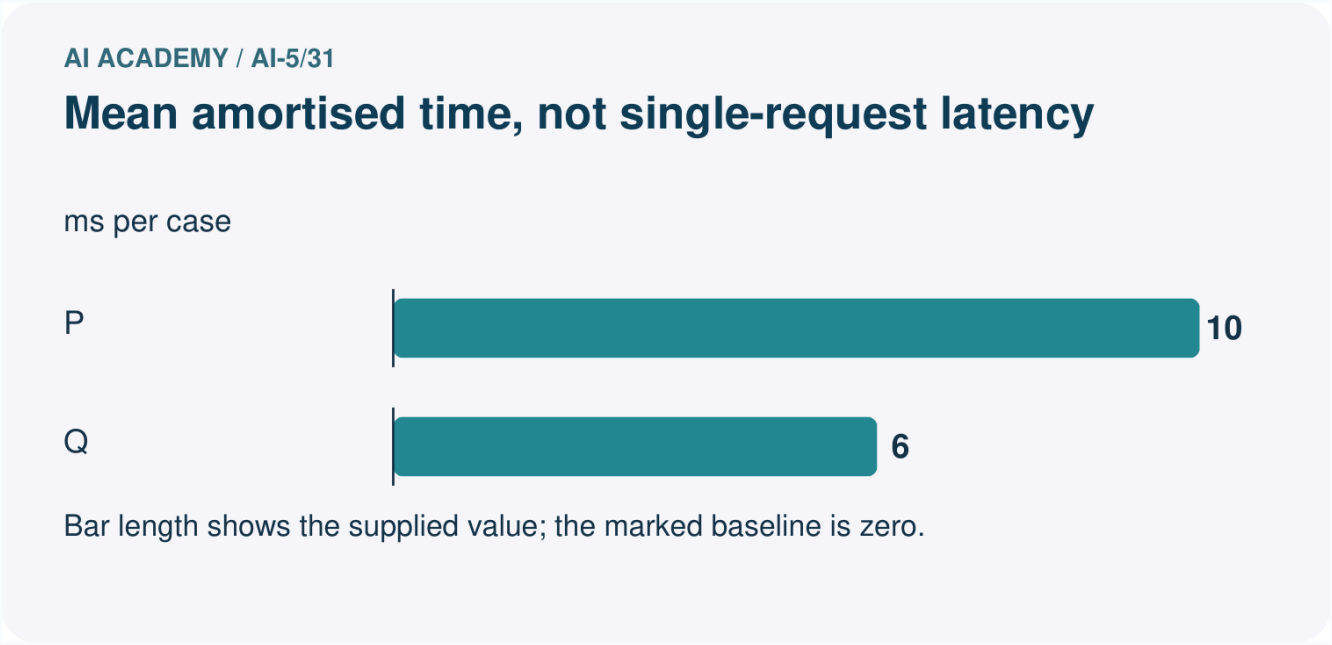
Complete task and evidence

- Compare speed and quality under the same case count and acceptance contract.

- Both evaluate 40 cases. P times [0.32, 0.40, 0.48]seconds, 38 correct; Q times [0.20, 0.24, 0.28]seconds, 36 correct. Requireaccuracy>=93% and mean amortised time<=12 ms/case.

- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Why is 10 ms/case not necessarily a single-request **latency**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/31

Efficiency and resource measurement

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Summarise P

Mean time 0.40s; 10 ms/case; accuracy 38/40=95%: passes both.

2. Summarise Q

Mean time 0.24s; 6 ms/case; accuracy 36/40=90%: fails quality.

3. Choose eligible candidate

P meets the declared contract; Q is faster but ineligible.

CHANGE ONE THING

Lower the accuracy requirement to90% before a new selection exercise.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Compare speed and quality under the same case count and acceptance contract.
- Which labelled step matches this detail? P meets the declared contract; Q is faster but ineligible.
- 1. Summarise P
- 2. Summarise Q
- 3. Choose eligible candidate
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Choose eligible candidate P meets the declared contract; Q is faster but ineligible. Compare models on the same workload and apply every required condition.

P meets both quality and time limits; Q's speed does not compensate for missing the quality gate. Dividing total run duration by forty gives an amortised time per case.

It does not measure the wait for one isolated request, which needs its own timed experiment.

Changed-case question and optional hint

- Changed-case practice: Lower the accuracy requirement to90% before a new selection exercise. Which conclusion is supported by the supplied facts?
- 1. Both become eligible; Q has the smaller measured mean amortised time.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Both become eligible; Q has the smaller measured mean amortised time. The acceptance contract changed; Q's measured accuracy did not improve.

Repair and retain the record

Review Dataset shift in neural models (ai-5/30). Goal: Explain how a changed **input** mix can change overall accuracy. Short practice: Did the supplied model's dim-light accuracy deteriorate?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Transfer learning (ai-5/28). Goal: Trace a reused feature extractor and a newly fitted **output** head. Short practice: Which part was trained for the new task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Sequence representations (ai-5/25). Goal: Encode an ordered sequence with an explicit vocabulary. Short practice: Which record lets the program distinguish the first and last zeros?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Efficiency and resource measurement: the case

AI ACADEMY / AI-5/31 · 01 · LOOK AT THE FACTS

Efficiency and resource measurement: the case

The situation

Compare speed and quality under the same case count and acceptance contract.

Supplied evidence

Both evaluate 40 cases. P times [0.32, 0.40, 0.48]seconds, 38 correct; Q times [0.20, 0.24, 0.28]seconds, 36 correct. Requireaccuracy>=93% and mean amortised time<=12 ms/case.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain efficiency and resource measurement using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

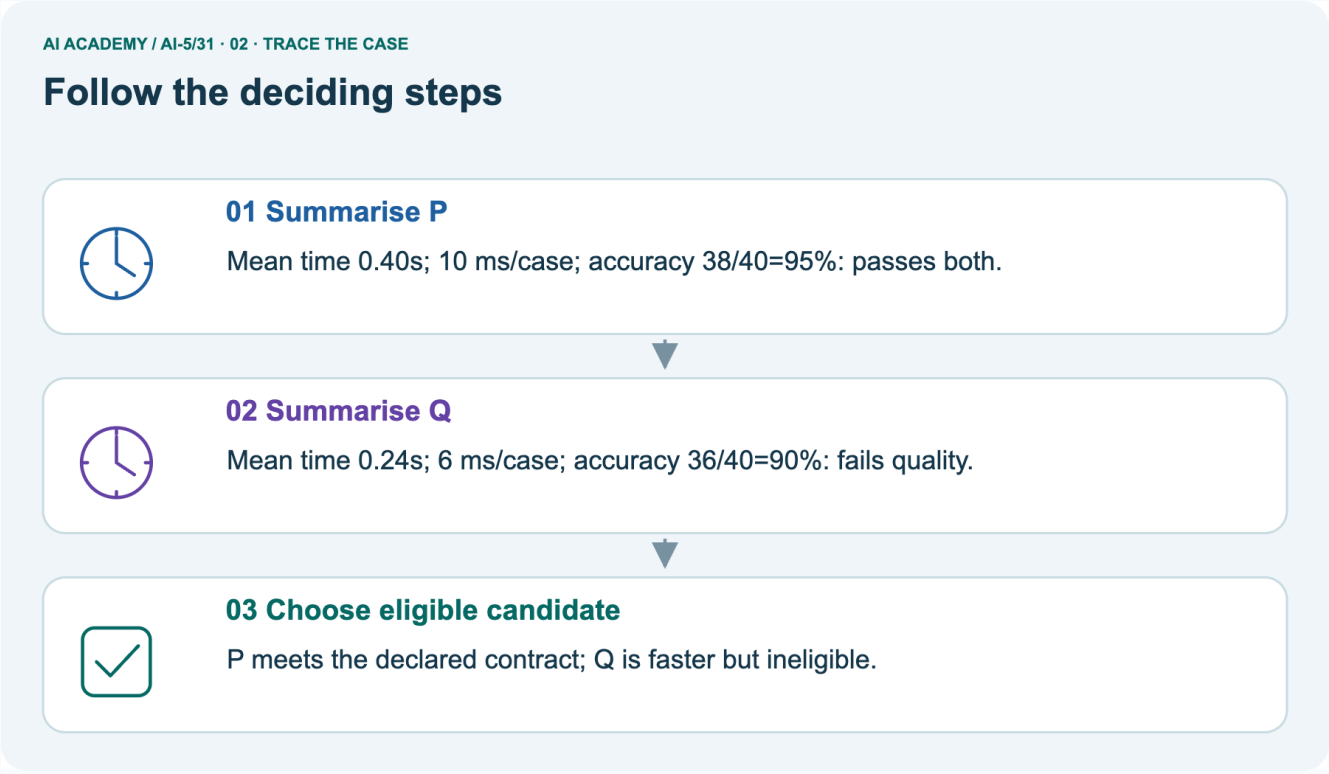
Compare speed and quality under the same case count and acceptance contract. Supplied facts: Both evaluate 40 cases. P times [0.32, 0.40, 0.48]seconds, 38 correct; Q times [0.20, 0.24, 0.28]seconds, 36 correct.

Requireaccuracy>=93% and mean amortised time<=12 ms/case. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



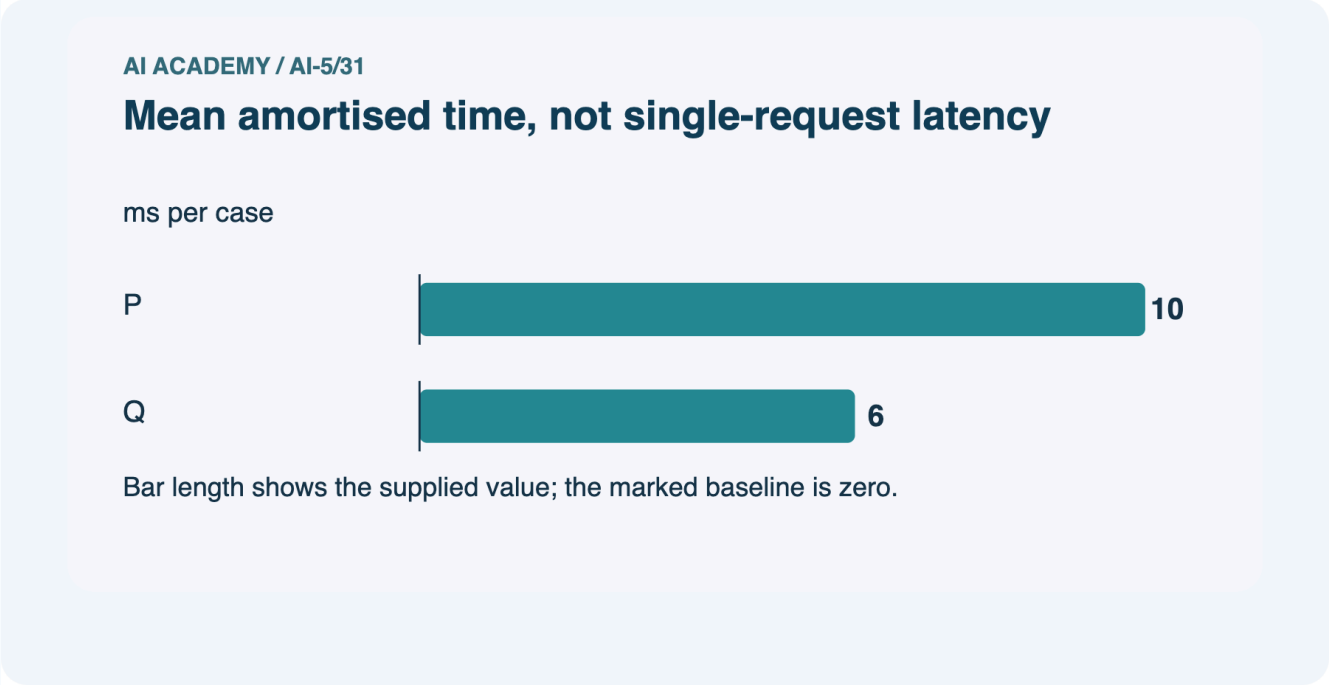
Follow the labelled picture with the matching explanation. Open the vector diagram

Summarise P: Mean time 0.40s; 10 ms/case; accuracy 38/40=95%: passes both. → Summarise Q: Mean time 0.24s; 6 ms/case; accuracy 36/40=90%: fails quality. → Choose eligible candidate: P meets the declared contract; Q is faster but ineligible.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Mean amortised time, not single-request latency



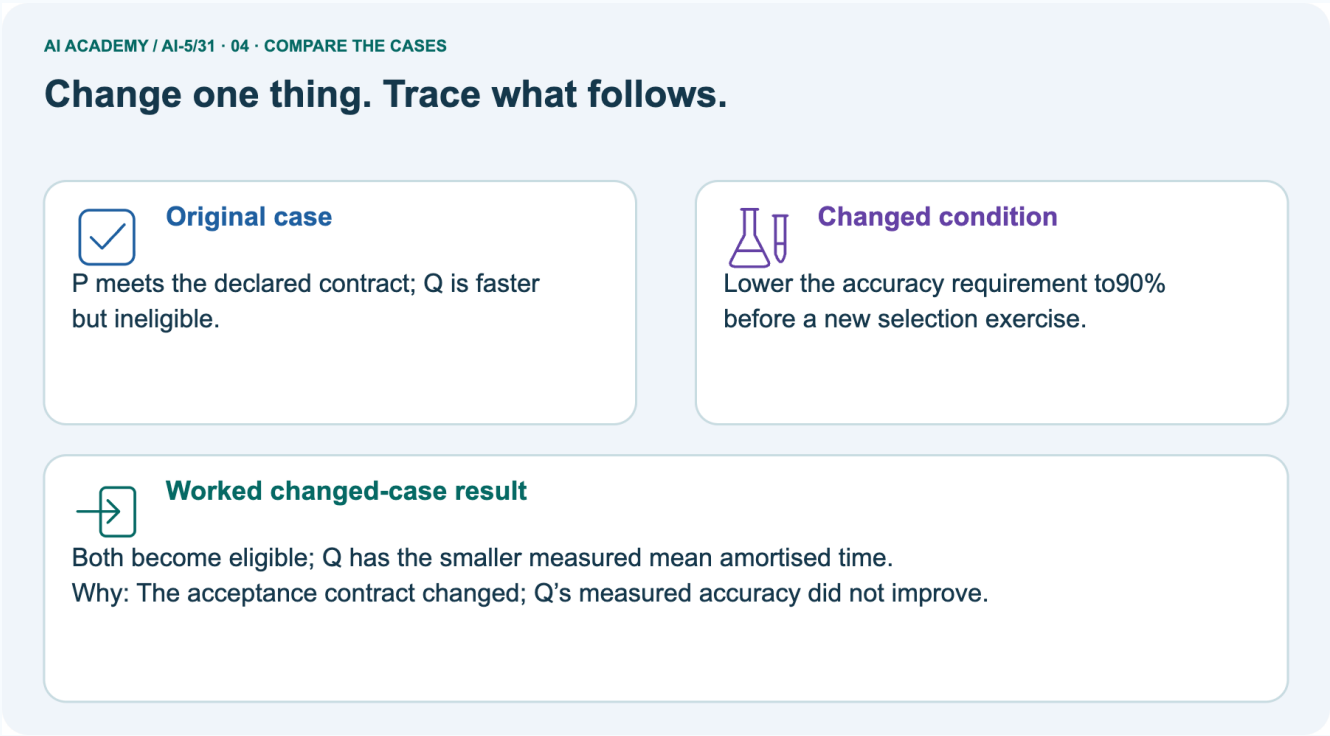
Follow the labelled picture with the matching explanation. Open the vector diagram

Mean amortised time, not single-request **latency**. Units: ms per case. P: 10. Q: 6.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

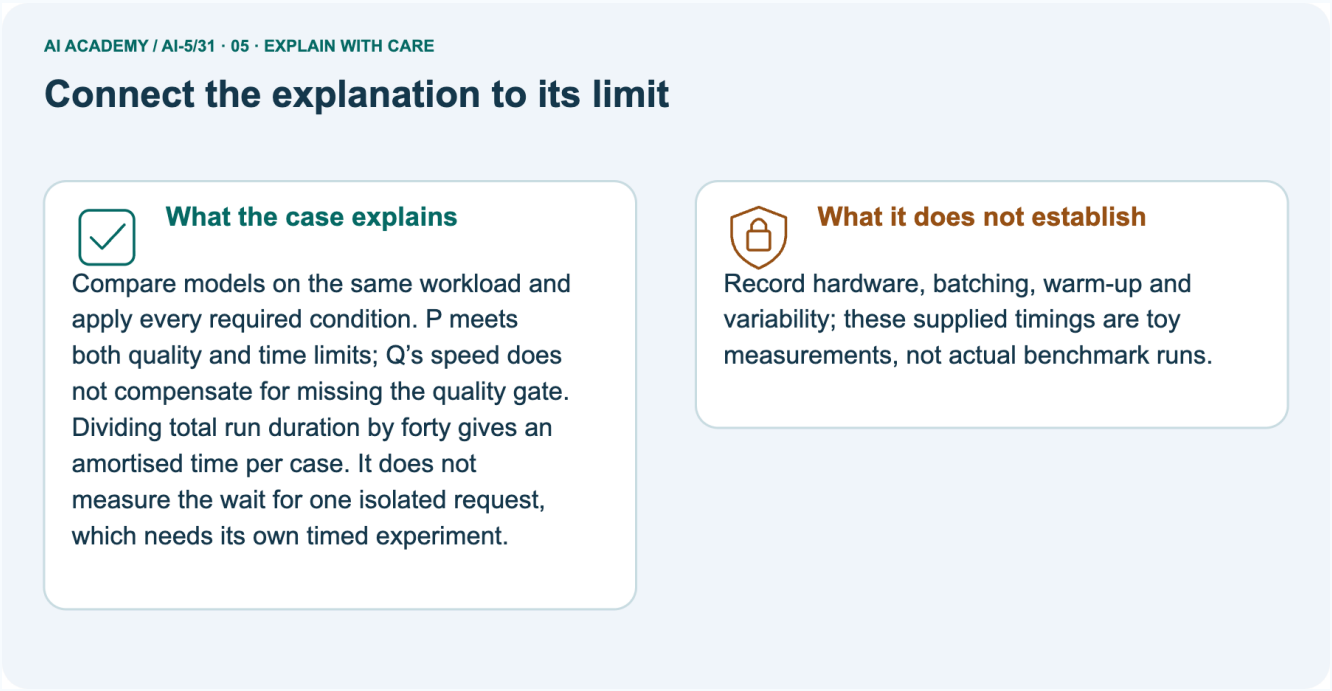
Original case: P meets the declared contract; Q is faster but ineligible. Changed condition: Lower the accuracy requirement to90% before a new selection exercise.

Expected result: Both become eligible; Q has the smaller measured mean amortised time. Reason: The acceptance contract changed; Q's measured accuracy did not improve.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare models on the same workload and apply every required condition. P meets both quality and time limits; Q's speed does not compensate for missing the quality gate.

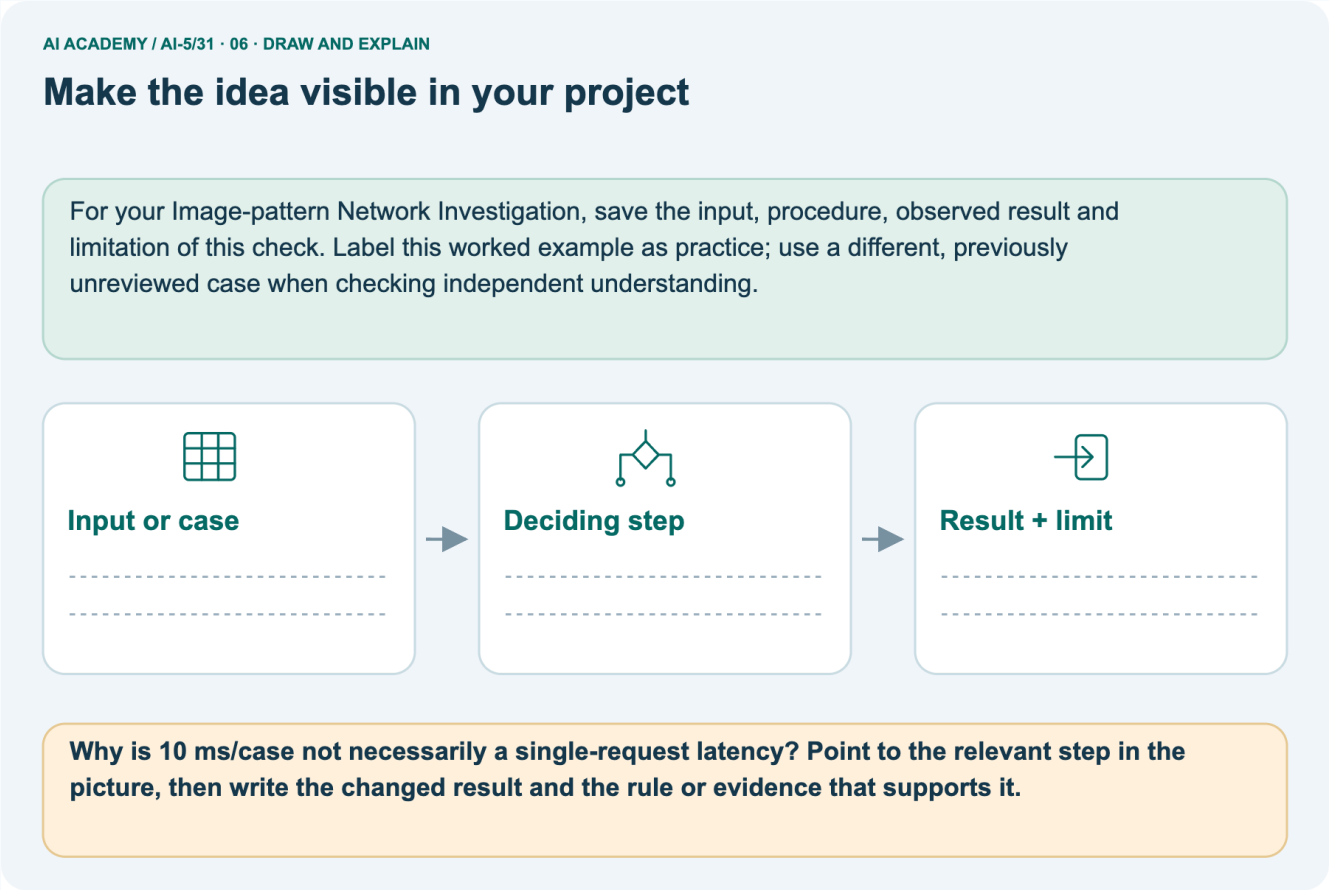
Dividing total run duration by forty gives an amortised time per case. It does not measure the wait for one isolated request, which needs its own timed experiment.

Limit: Record hardware, batching, warm-up and variability; these supplied timings are toy measurements, not actual benchmark runs.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is 10 ms/case not necessarily a single-request **latency**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Efficiency and resource measurement: the case

02. Follow the deciding steps

03. Mean amortised time, not single-request latency

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.